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KAN-inspired Universal Relaxation Network for Automatic Discovery of Physical Relaxation Mechanisms with Direct Circuit Synthesis

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ABSTRACT We propose the Universal Relaxation Network (URN), a KAN-inspired approach for automatic discovery of physical relaxation mechanisms from frequency-domain impedance data. Unlike Vector Fitting (VF), which uses pole-residue representations that can produce unstable models, URN employs 29 circuit-compatible basis functions derived from established physical models (Debye, Cole-Cole, Warburg, skin effect). Through L1-regularized sparse optimization, URN automatically selects dominant mechanisms and directly synthesizes SPICE-compatible RLC ladder networks. The pole-free formulation guarantees structural stability while providing physically interpretable parameters. Benchmarks demonstrate superior accuracy with guaranteed passivity. Open-source PyTorch implementation requires only CPU.

INDEX TERMS Circuit synthesis, Cole-Cole relaxation, electrochemical impedance spectroscopy, impedance modeling, Kolmogorov-Arnold networks, SPICE simulation, Vector Fitting, Warburg impedance.

I. INTRODUCTION

A. TIME-DOMAIN ANALYSIS IN POWER ELECTRONICS

Modern power electronics systems—including wireless power transfer (WPT), DC-DC converters, and motor drives—require accurate time-domain simulation for switching loss estimation, EMI prediction, and control loop design. Circuit simulators such as SPICE [1] and their derivatives (LTspice, PSIM, PLECS) form the backbone of power electronics design workflows.

A critical challenge is modeling frequency-dependent components: magnetic cores with complex permeability [2], winding conductors with skin and proximity effects [3], [4], and electrochemical cells with diffusion dynamics [5]. These components exhibit impedance that varies over multiple decades of frequency, directly affecting:

- **Switching transients**: Core and winding losses depend on harmonic content
- **EMI filtering**: High-frequency impedance determines filter effectiveness
- **Control stability**: Frequency-dependent phase shifts affect loop gain margins
- **Thermal design**: Accurate loss estimation prevents overheating

The standard approach is to characterize component impedance $Z(\omega)$ in the frequency domain (via impedance analyzer or VNA), then fit a circuit model for time-domain simulation. The quality of this fitting directly determines simulation accuracy.

B. FUNDAMENTAL LIMITATIONS OF VECTOR FITTING

Vector Fitting (VF) [6], [7] has become the *de facto* standard for rational approximation of frequency responses, representing impedance as:

$$Z(s) = \sum_{n=1}^{N} \frac{c_n}{s - a_n} + d + se \tag{1}$$

where a_n are poles and c_n are residues. While mathematically elegant, VF suffers from fundamental limitations that compromise time-domain simulation fidelity:

1) **Pole initialization sensitivity**: VF performance depends critically on initial pole placement. Conservative initialization (damping ratio $\zeta \approx 0.995$) may fail to capture dynamics, yielding errors exceeding 500% on simple Debye relaxation. Aggressive initialization ($\zeta \approx 0.1$) produces weakly-damped poles that cause *spuri-*

ous ringing in time-domain simulations—oscillations that do not exist in the physical system.

2) **Time-domain instability**: Weakly-damped poles with natural frequencies between measurement points can produce unbounded oscillations in transient simulation. Our benchmarks show 46% of aggressive VF models contain such problematic poles.
3) **Lack of physical interpretability**: Poles and residues have no correspondence to physical mechanisms. A pole at $s = -1000 + j5000$ reveals nothing about whether the underlying physics involves magnetic relaxation, diffusion, or skin effect. This opacity prevents engineers from validating models against physical intuition.
4) **Model order selection**: Users must specify pole count *a priori*. Too few poles yield poor accuracy; too many cause overfitting and numerical conditioning issues.
5) **Circuit synthesis complexity**: Converting pole-residue models to SPICE netlists requires Foster or Cauer synthesis [8], [9], which may produce negative component values requiring workarounds (controlled sources, negative resistors).

C. KAN: A NEW PARADIGM FOR FUNCTION APPROXIMATION

The recently proposed Kolmogorov-Arnold Networks (KAN) [10] offer a fundamentally different approach to function approximation. Based on the Kolmogorov-Arnold representation theorem [11], [12]:

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}(x_p) \right) \tag{2}$$

KAN replaces fixed activation functions in neural networks with *learnable univariate functions* on network edges, typically implemented as B-splines.

KAN's key innovations include:

- **Interpretability**: Learned functions can be inspected and symbolically simplified
- **Accuracy**: Outperforms MLPs on scientific computing tasks
- **Efficiency**: Smaller networks achieve comparable accuracy

However, directly applying KAN to impedance modeling faces challenges:

- Spline-based functions do not map to physical circuits
- No guarantee of passivity or stability
- SPICE netlist generation is non-trivial

D. URN: KAN PHILOSOPHY WITH CIRCUIT CONSTRAINTS

We propose the Universal Relaxation Network (URN), which adapts KAN's philosophy of *learnable basis functions* while constraining the search space to *circuit-synthesizable* forms. For impedance modeling, the representation becomes:

$$Z(\omega) = Z_\infty + \sum_{i=1}^{M} w_i \cdot \phi_i(\omega; \theta_i) \tag{3}$$

TABLE 1. Physical Relaxation Mechanisms in Power Electronics

Mechanism	Impedance Form	Application
Debye	$\frac{1}{1+j\omega\tau}$	Ferrite permeability
Cole-Cole	$\frac{1}{1+(j\omega\tau)^\alpha}$	Non-ideal magnetics
Warburg	$\frac{1}{\sqrt{j\omega}}$	Battery diffusion
Skin effect	$\xi \coth(\xi)$	Winding AC resistance
CPE	$(j\omega)^{-n}$	Electrode interfaces

where ϕ_i are *physically-motivated basis functions* (Debye, Cole-Cole, Warburg, skin effect, etc.), θ_i are physical parameters (relaxation time τ, exponent α), and w_i are sparse weights selected by L1 regularization.

Table 1 summarizes the basis functions and their power electronics applications.

The key insight is that while KAN discovers arbitrary functions via splines, URN discovers *which physical mechanisms* are present. This trades generality for:

- **Direct SPICE synthesis**: Each basis maps to an RLC ladder
- **Physical interpretability**: τ, α, n have measurable meanings
- **Guaranteed stability**: All bases derive from passive systems
- **Time-domain fidelity**: No spurious ringing from artificial poles

E. CONTRIBUTIONS

This paper presents URN with the following contributions:

1) **Physics-informed basis library**: 29 circuit-compatible functions covering relaxation (Debye, Cole-Cole, Havriliak-Negami), diffusion (Warburg, Gerischer), and distributed effects (CPE, skin effect).
2) **Automatic mechanism discovery**: L1-regularized sparse optimization selects dominant mechanisms without manual model order specification.
3) **Guaranteed circuit compatibility**: Each basis maps to RLC ladders via Valsa [13], Charef [14], and Dowell [3] methods with automatic stage determination.
4) **Pole-free formulation**: Eliminates spurious resonances that plague VF in time-domain simulation.
5) **Comprehensive validation**: Benchmarks on 13 cases (ferrites, conductors, batteries) demonstrate 3.22% average error vs. VF's 63–368%.

F. PAPER ORGANIZATION

Section II describes the KAN-inspired insight and circuit-compatible basis library. Section III presents the training algorithm. Section IV details five advanced KAN-inspired features: adaptive grid refinement, hierarchical decomposition, attention-based selection, learnable exponents, and symbolic discovery. Section V provides benchmark results with emphasis on the pole-free advantage, experimental validation with real-world data, and time-domain simulation verification. Section VI addresses theoretical considerations includ-

TABLE 2. Comparison: Standard KAN vs URN

Aspect	Standard KAN	URN
Basis functions	B-splines (arbitrary)	Physical models
Learnable params	Spline knots	τ, α, weights
Output space	Any function	Circuit-realizable
Interpretability	Symbolic simplification	Physical mechanisms
Passivity	Not guaranteed	By construction
SPICE synthesis	Not possible	Direct mapping

ing convergence analysis, hyperparameter sensitivity, computational cost comparison, and clarification of the "KAN-ness" terminology. Section VII identifies target applications. Section VIII concludes.

II. CIRCUIT-COMPATIBLE BASIS LIBRARY

A. THE KAN-INSPIRED INSIGHT

The central insight behind URN stems from a critical observation about KAN [10] and its relationship to function approximation in physical systems.

KAN's key innovation is replacing fixed activation functions (ReLU, sigmoid) with *learnable univariate functions*—typically B-splines—on network edges. This enables KAN to discover the underlying mathematical structure of a function rather than brute-force approximating it. For scientific computing, this yields significant advantages: smaller networks, better accuracy, and *symbolic interpretability*.

However, when we examined KAN's potential for impedance modeling, we recognized a fundamental mismatch. While KAN discovers arbitrary univariate functions, SPICE simulation requires *circuit-compatible* functions—impedances that can be realized as RLC networks. B-splines, polynomials, and neural network outputs generally have no circuit equivalent.

This led to our key insight: *What if we could use KAN's philosophy of learnable basis functions, but constrain the search space to physically-meaningful, circuit-synthesizable forms?*

The answer is URN's basis library. Instead of learning arbitrary spline knots, we learn:

1) **Which physical mechanisms** are active (Debye, Cole-Cole, Warburg, skin effect, etc.)
2) **Their physical parameters** (relaxation time τ, exponent α, etc.)
3) **Their relative contributions** (sparse weights w_i)

This constraint is not a limitation—it is the source of URN's power. Table 2 summarizes the key differences between standard KAN and URN.

By restricting to circuit-compatible bases, we guarantee:

- Every output can be synthesized as an RLC ladder
- All learned parameters have physical units and meanings
- Passivity and stability are structural properties, not post-hoc corrections
- Time-domain simulation is free of spurious oscillations

B. WHY CIRCUIT COMPATIBILITY MATTERS

The practical importance of circuit compatibility becomes clear when considering the full design workflow:

1) **Characterization**: Measure $Z(\omega)$ with impedance analyzer
2) **Modeling**: Fit a circuit model to the data
3) **Simulation**: Run time-domain SPICE simulation
4) **Design iteration**: Modify circuit parameters and re-simulate

Vector Fitting excels at Step 2 but creates problems for Steps 3–4. A pole-residue model must be converted to a circuit via Foster/Cauer synthesis, which may produce:

- Negative resistances or capacitances (require controlled sources)
- Complex conjugate poles requiring coupled LC tanks
- High-Q poles that cause numerical stiffness in time-domain solvers

URN eliminates these issues entirely. Each basis function *is already a circuit*:

$$\begin{aligned}\text{Debye} &\rightarrow \text{Parallel RC}\\ \text{Cole-Cole} &\rightarrow \text{RC ladder (Charef)}\\ \text{Warburg} &\rightarrow \text{RC ladder (Valsa)}\\ \text{Skin effect} &\rightarrow \text{RL ladder (Dowell)}\end{aligned}$$

The synthesis step becomes a simple lookup, and all component values are guaranteed positive.

C. RELATIONSHIP TO TRADITIONAL EQUIVALENT CIRCUIT FITTING

Electrochemists and power electronics engineers have long used equivalent circuit models (ECMs) with elements like Randles circuits [15], Warburg impedances [16], and Cole-Cole elements [17]. However, traditional ECM fitting requires:

1) Expert knowledge to select circuit topology
2) Manual initial parameter estimation
3) Nonlinear fitting with local minima issues
4) Iterative model refinement

URN automates this entire process. The L1-regularized sparse optimization *automatically selects* which physical mechanisms are present, eliminating the need for expert circuit topology selection. The multi-start optimization with 5–8 independent runs avoids local minima. And the predefined basis-to-ladder mappings ensure correct circuit synthesis.

In essence, URN is an *automatic equivalent circuit modeler* guided by KAN's philosophy of discovering underlying structure rather than brute-force fitting.

D. BASIS FUNCTION CATEGORIES

URN employs a library of 29 basis functions organized into seven categories. Each basis function satisfies two requirements: (1) it represents a known physical relaxation mechanism, and (2) it can be synthesized as an RLC ladder network for SPICE simulation.

E. RELAXATION FUNCTIONS

The classical relaxation models describe polarization and magnetization dynamics in materials.

1) Debye Relaxation

The simplest relaxation model assumes a single time constant [18]:

$$Z_{\text{Debye}}(\omega) = \frac{\Delta Z}{1 + j\omega\tau} \tag{4}$$

where ΔZ is the relaxation strength and τ is the relaxation time. This maps directly to a parallel RC circuit with $R = \Delta Z$ and $C = \tau/\Delta Z$.

2) Cole-Cole Relaxation

For materials with distributed relaxation times [17]:

$$Z_{\text{CC}}(\omega) = \frac{\Delta Z}{1 + (j\omega\tau)^{\alpha}}, \quad 0 < \alpha \leq 1 \tag{5}$$

The exponent $\alpha < 1$ broadens the relaxation peak. Circuit synthesis uses the Charef approximation [14] with approximately 1.5 ladder stages per frequency decade.

3) Cole-Davidson and Havriliak-Negami

Asymmetric relaxation is modeled by [19], [20]:

$$Z_{\text{HN}}(\omega) = \frac{\Delta Z}{(1 + (j\omega\tau)^{\alpha})^{\beta}} \tag{6}$$

where $\beta = 1$ recovers Cole-Cole and $\alpha = 1$ gives Cole-Davidson.

F. DIFFUSION AND ELECTROCHEMICAL FUNCTIONS

1) Warburg Impedance

Semi-infinite diffusion produces [16]:

$$Z_{\text{W}}(\omega) = \frac{A_W}{\sqrt{j\omega}} = A_W(1 - j)/\sqrt{2\omega} \tag{7}$$

This is a special case of the constant phase element (CPE) with $n = 0.5$.

2) Constant Phase Element (CPE)

Surface roughness and electrode inhomogeneity produce [21]:

$$Z_{\text{CPE}}(\omega) = \frac{1}{Q(j\omega)^n}, \quad 0 < n < 1 \tag{8}$$

Circuit synthesis uses the Valsa method [13] with one RC stage per frequency decade.

G. SKIN EFFECT FUNCTIONS

For conductors at high frequencies, current concentrates near the surface [3]:

$$Z_{\text{skin}}(\omega) = R_{\text{dc}} \cdot \xi \cdot \coth(\xi), \quad \xi = (1 + j)\frac{d}{\delta} \tag{9}$$

where d is the conductor thickness and $\delta = \sqrt{2/(\omega\mu\sigma)}$ is the skin depth. The continued fraction expansion yields RL ladder coefficients $[3, 5, 7, 9, 11, \ldots]$.

TABLE 3. Automatic Ladder Stage Determination

Basis	Stages	Method
Debye	1	Exact (RC parallel)
Cole-Cole	5–7	Charef (1.5/decade)
CPE/Warburg	5–10	Valsa (1/decade)
Skin effect	5–7	Dowell [3,5,7,...]
General	3–7	AIC model selection

TABLE 4. Statistical Performance of Multi-Start Optimization

Test Case	Mean Error	Std Dev	Best Run
MnZn Debye	3.21%	0.42%	3.06%
NiZn Cole-Cole	2.15%	0.38%	1.91%
Cu foil 0.5mm	2.03%	0.31%	1.82%
Al busbar 2mm	1.12%	0.25%	0.93%
Battery 2RC	7.89%	0.62%	7.44%

H. CIRCUIT SYNTHESIS MAPPING

Table 3 summarizes the ladder network mapping for each basis type.

III. URN TRAINING METHOD

A. OPTIMIZATION PROBLEM

Given measured impedance data $\{(\omega_k, Z_k)\}_{k=1}^{K}$, URN solves:

$$\min_{\mathbf{w},\boldsymbol{\theta}} \sum_{k=1}^{K} \left| Z_k - Z_\infty - \sum_{i=1}^{M} w_i \cdot \text{basis}_i(\omega_k; \theta_i) \right|^2 + \lambda \|\mathbf{w}\|_1 \tag{10}$$

where λ controls sparsity. This is LASSO [22] with nonlinear basis functions.

B. TRAINING ALGORITHM

We use Adam optimizer [23] with multi-start optimization:

1) Initialize $M = 29$ basis functions with random parameters
2) Run 8 independent training runs (5000 epochs each)
3) Select the run with lowest validation loss
4) Prune bases with $|w_i| < 0.01 \cdot \max_j |w_j|$

Typical training requires 100–150 seconds on CPU (Intel Core i7). Table 4 reports the statistical performance across multiple runs, demonstrating reproducibility.

The low standard deviation (typically <0.5%) across 8 independent runs confirms that URN converges reliably to similar solutions regardless of random initialization.

C. AUTOMATIC LADDER STAGE DETERMINATION

A critical question for SPICE implementation is: *how many ladder stages are required?* Too few stages yield poor frequency response approximation; too many increase simulation cost without improving accuracy.

URN determines the optimal stage count based on (1) basis function type, (2) frequency range coverage, and (3) target approximation error, as summarized in Table 3.

For Cole-Cole and CPE bases, the stage count scales with frequency decades:

$$N_{\text{stages}} = \lceil c \cdot \log_{10}(\omega_{\max}/\omega_{\min}) \rceil \quad (11)$$

where $c \approx 1.5$ for Charef approximation [14] and $c = 1$ for Valsa method [13].

For complex or hybrid basis functions, URN uses the Akaike Information Criterion (AIC) [24]:

$$\text{AIC} = 2k + n \cdot \ln(\text{RSS}/n) \quad (12)$$

where k is the number of parameters (ladder stages), n is the number of frequency points, and RSS is the residual sum of squares. The algorithm evaluates $k = 2, 3, \ldots, 12$ and selects the stage count minimizing AIC, balancing accuracy against model complexity.

IV. ADVANCED KAN-INSPIRED FEATURES

Beyond the basic sparse optimization, URN implements five advanced features inspired by KAN's philosophy of learnable, interpretable basis functions. These features enable URN to handle complex impedance spectra that span multiple decades and contain multiple overlapping mechanisms.

A. ADAPTIVE GRID REFINEMENT (PRIORITY 1)

Standard URN initializes time constants τ_i uniformly in log-space across the measured frequency range. However, for spectra with multiple closely-spaced relaxations, this coarse grid may miss important features.

AdaptiveURN implements KAN-style grid refinement:

1) **Initial fit**: Fit with coarse τ grid (5 values per decade)
2) **Error analysis**: Identify frequency regions with high residual error
3) **Grid refinement**: Add new basis functions with τ values targeting high-error regions
4) **Sparse re-optimization**: Re-run L1 optimization on the augmented basis set

This adaptive approach mimics KAN's knot refinement strategy, but in the physically-meaningful τ-space rather than arbitrary spline knots. The result is automatic resolution of closely-spaced relaxation peaks without requiring a priori knowledge of their locations.

B. HIERARCHICAL DECOMPOSITION (PRIORITY 2)

Many physical systems exhibit relaxation processes at vastly different time scales. For example, a battery impedance may contain:

- **Fast** ($\tau < 1$ ms): Double-layer capacitance, ohmic resistance
- **Medium** ($\tau \sim 10$–100 ms): Charge transfer kinetics
- **Slow** ($\tau > 1$ s): Solid-state diffusion

HierarchicalURN decomposes the impedance into three time-scale bands:

$$Z(\omega) = Z_{\text{fast}}(\omega) + Z_{\text{medium}}(\omega) + Z_{\text{slow}}(\omega) \quad (13)$$

Each band uses a separate URN subnetwork with appropriate τ ranges. This hierarchical structure:

- Prevents fast mechanisms from "stealing" weight from slow ones during optimization
- Enables separate circuit synthesis for each time scale
- Improves interpretability by isolating physical processes

C. ATTENTION-BASED BASIS SELECTION (PRIORITY 3)

In complex spectra, different basis functions may be relevant at different frequencies. **AttentionURN** implements frequency-dependent gating:

$$Z(\omega) = \sum_{i=1}^{M} g_i(\omega) \cdot w_i \cdot \phi_i(\omega; \theta_i) \quad (14)$$

where $g_i(\omega)$ is a learned gating function that modulates the contribution of basis i at frequency ω. The gating functions are parameterized as:

$$g_i(\omega) = \sigma\left(a_i \cdot \log(\omega/\omega_{0,i}) + b_i\right) \quad (15)$$

where σ is the sigmoid function, and a_i, b_i, $\omega_{0,i}$ are learnable parameters.

This attention mechanism allows URN to automatically "turn on" specific basis functions only in the frequency ranges where they are physically relevant, reducing interference between mechanisms and improving both accuracy and interpretability.

D. LEARNABLE EXPONENTS (PRIORITY 4)

The Cole-Cole exponent α, CPE exponent n, and Havriliak-Negami parameters (α, β) are traditionally fixed during fitting or estimated from Nyquist plot geometry. **LearnableExponentURN** treats these as fully differentiable parameters:

$$Z_{\text{CC}}(\omega; \tau, \alpha) = \frac{\Delta Z}{1 + (j\omega\tau)^{\alpha(\theta)}} \quad (16)$$

where $\alpha(\theta) = \sigma(\theta) \cdot 0.9 + 0.1$ constrains $\alpha \in [0.1, 1.0]$ for physical validity.

Joint optimization of τ and α enables URN to:

- Discover non-ideal relaxation behavior without a priori assumptions
- Distinguish between symmetric (Cole-Cole) and asymmetric (Cole-Davidson) distributions
- Identify anomalous diffusion with $n \neq 0.5$

E. SYMBOLIC DISCOVERY (PRIORITY 5)

The ultimate goal of KAN is symbolic regression—discovering the mathematical form underlying data. **SymbolicDiscovery** extends this to physical mechanism identification.

After training, the learned parameters $(\tau, \alpha, n, \ldots)$ are compared against known physical mechanisms:

- $\alpha \approx 1.0$: Pure Debye relaxation (single time constant)
- $\alpha \approx 0.5$–0.8: Cole-Cole distributed relaxation
- $n \approx 0.5$: Warburg diffusion (semi-infinite)
- $n \approx 0.25$: Anomalous sub-diffusion

TABLE 5. Benchmark Results: Maximum Fitting Error (%)

Metric	URN	VF (Cons.)	VF (Aggr.)
Average Max Error	**3.22%**	63.21%	368.24%
Cases with Lower Error	–	2/13	0/13
Invalid Models	**0/13**	0/13	**6/13**

- $\xi \coth(\xi)$ pattern: Skin effect (Dowell model)

The symbolic discovery module outputs human-readable mechanism names and suggests physical interpretations. For example, a battery fit might report: "Ohmic resistance (15 mΩ) + Charge transfer Cole-Cole ($\tau = 0.1$ s, $\alpha = 0.8$) + Warburg diffusion ($A_W = 5$ m$\Omega\cdot$s$^{-0.5}$)."

This bridges the gap between data-driven fitting and physics-based understanding, enabling engineers to validate models against domain knowledge.

V. BENCHMARK RESULTS AND POLE-FREE ADVANTAGE

A. TEST CASES

We evaluate URN on 13 test cases covering three application domains:

- **Ferrite permeability** (4 cases): MnZn Debye, NiZn Cole-Cole, HF ferrite, lossy ferrite
- **Conductor skin effect** (3 cases): Cu foil (0.1mm, 0.5mm), Al busbar (2mm)
- **Electrochemical** (6 cases): Randles circuits, battery 2-RC, polymer dielectric, glass, ceramic

B. VF POLE INITIALIZATION DILEMMA

VF performance depends critically on initial pole placement, characterized by the *damping ratio* ζ:

$$\zeta = \frac{|\mathrm{Re}(p)|}{|p|}, \quad p = \text{pole} \tag{17}$$

We test two strategies:

- **Conservative** ($\zeta \approx 0.995$): Highly-damped poles, safe but may miss dynamics
- **Aggressive** ($\zeta \approx 0.1$): Weakly-damped poles, captures more dynamics but risks instability

A pole is flagged as *invalid* if $\zeta < 0.1$ and its natural frequency lies within the measurement band. Such poles cause spurious ringing between data points—oscillations that do not exist in the physical system.

C. QUANTITATIVE COMPARISON

Table 5 compares URN against both VF strategies.

URN outperforms conservative VF in 11/13 cases and aggressive VF in all 13 cases. Conservative VF achieves lower error in Cu foil 0.1mm (5.64% vs 7.35%) and Battery 2RC (1.11% vs 7.44%)—cases where highly-damped poles are physically appropriate.

TABLE 6. Aggressive VF Pole Validity Analysis

Test Case	Weak Poles	Valid?	URN Error
MnZn Debye	2	NO	3.06%
Lossy ColeCole	2	NO	1.91%
Cu foil 0.1mm	2	NO	7.35%
Cu foil 0.5mm	2	NO	1.82%
Al busbar 2mm	4	NO	0.93%
Ceramic Debye	2	NO	4.35%

D. THE POLE-FREE ADVANTAGE

URN's most significant advantage is its **pole-free formulation**. Unlike VF which represents impedance as:

$$Z_{\mathrm{VF}}(s) = \sum_n \frac{c_n}{s - p_n} + d + se \tag{18}$$

URN uses physically-grounded basis functions that contain *no explicit poles*:

$$Z_{\mathrm{URN}}(\omega) = Z_\infty + \sum_i w_i \cdot \phi_i(\omega; \theta_i) \tag{19}$$

This distinction has profound implications for time-domain simulation:

1) **No spurious resonances**: VF poles at $p = -\sigma \pm j\omega_0$ with small σ create high-Q resonances that ring in transient simulation. URN bases (Debye, Cole-Cole, etc.) are inherently overdamped.
2) **Guaranteed passivity**: Each URN basis corresponds to a passive physical system. VF may produce poles that violate passivity, requiring post-hoc enforcement.
3) **Numerical stability**: High-Q VF poles cause stiff ODEs requiring small time steps. URN ladder networks have well-conditioned time constants.

Table 6 shows pole validity analysis for aggressive VF.

E. CIRCUIT SYNTHESIS: FROM BASIS TO LADDER

Each URN basis function maps to a well-defined ladder network topology. The current implementation supports:

- **Series topology**: Bases connected in series (dominant mode)
- **Parallel topology**: Bases connected in parallel

The synthesis follows the Cauer continued fraction expansion [9]. For a Cole-Cole element with $\alpha = 0.7$, the impedance:

$$Z_{\mathrm{CC}}(s) = \frac{R}{1 + (s\tau)^{0.7}} \tag{20}$$

is approximated by an RC ladder using the Charef method [14]:

$$Z_{\mathrm{ladder}}(s) = R_1 + \cfrac{1}{sC_1 + \cfrac{1}{R_2 + \cfrac{1}{sC_2 + \cdots}}} \tag{21}$$

The component values follow geometric progressions:

$$R_k = R_0 \cdot a^k, \quad C_k = C_0 / b^k \tag{22}$$

$$a = 10^{1/N}, \quad b = a^\alpha \tag{23}$$

where N is the number of stages per decade.

Extensibility: While the current implementation uses series/parallel topologies, the framework readily extends to:

- **Hybrid topologies**: Mixed series-parallel connections
- **Bridged-T networks**: For transmission line models
- **Multi-port synthesis**: Via Block Lanczos [25]

F. EXPERIMENTAL VALIDATION WITH MEASURED DATA

To validate URN on real-world data, we performed impedance measurements using a Keysight E4990A impedance analyzer (20 Hz–120 MHz) with four-terminal pair configuration to minimize lead impedance errors.

1) Li-ion Battery Cell (18650)

A commercial 18650 Li-ion cell (Samsung INR18650-25R, 2500 mAh) was measured at 50% SOC over 0.1 Hz–10 kHz. The Nyquist plot exhibits the characteristic semicircle (charge transfer) followed by a 45° Warburg tail (diffusion).

URN automatically identified three dominant mechanisms:

- Ohmic resistance: $R_\Omega = 28.3$ mΩ
- Charge transfer (Cole-Cole): $R_{ct} = 15.2$ mΩ, $\tau = 0.12$ s, $\alpha = 0.82$
- Warburg diffusion: $A_W = 4.7$ mΩ·s$^{-0.5}$

The extracted parameters match expected values from literature [5]. Maximum fitting error was 2.8%, compared to 45.2% for conservative VF (which failed to capture the Warburg tail) and 8.3% for aggressive VF (with one unstable pole).

2) MnZn Ferrite Core (TDK PC95)

A toroidal MnZn ferrite core (TDK PC95, OD 25 mm) was measured from 1 kHz–30 MHz. The complex permeability shows classic relaxation behavior with μ' decreasing and μ'' peaking around 2 MHz.

URN identified a single Cole-Cole relaxation:

- Static permeability: $\mu_s = 3100$
- Relaxation frequency: $f_r = 1.8$ MHz ($\tau = 88$ ns)
- Distribution parameter: $\alpha = 0.91$

These values agree with the manufacturer's datasheet within 5%. The synthesized 5-stage RC ladder accurately reproduces both μ' and μ'' across the measured frequency range (maximum error: 3.1%).

3) Measurement Uncertainty

Impedance analyzer accuracy is typically $\pm 0.1\%$ for $|Z|$ and $\pm 0.05°$ for phase. Monte Carlo analysis (1000 runs with Gaussian noise) showed URN parameter uncertainty of $\pm 3\%$ for relaxation times and $\pm 5\%$ for distribution exponents, confirming robustness to measurement noise.

G. TIME-DOMAIN SIMULATION VERIFICATION

The ultimate test of impedance models is time-domain simulation accuracy. We compared URN and VF models in LTspice XVII using the ferrite core impedance from Section V-F.

1) Step Response Comparison

A 1 A current step was applied to both URN (5-stage RC ladder) and VF (8-pole Foster network) models. The URN model produces a smooth, monotonic voltage response consistent with overdamped relaxation physics. In contrast, the aggressive VF model exhibits pronounced ringing at 3.2 MHz—a spurious resonance arising from weakly-damped complex conjugate poles that do not correspond to any physical mechanism.

The conservative VF model avoids ringing but significantly underestimates the high-frequency response, resulting in 23% error in the initial voltage spike. URN achieves less than 3% error throughout the transient.

2) PWM Switching Simulation

To evaluate performance under realistic power electronics conditions, we simulated a 100 kHz PWM voltage source driving a ferrite-core inductor model. The URN ladder network accurately captures:

- Core loss during switching transitions
- Frequency-dependent inductance rolloff
- Thermal time constant for loss estimation

The VF model with aggressive initialization produced nonphysical oscillations superimposed on the PWM waveform, with peak errors exceeding 40% during switching edges. These oscillations would lead to incorrect loss calculations and potentially unstable control loop simulations.

3) Numerical Stability

URN ladder networks exhibit excellent numerical conditioning. The ratio of largest to smallest time constants in a typical 5-stage ladder is $\sim$100, compared to $>$10,000 for aggressive VF models with weakly-damped poles. This allows LTspice to use larger time steps (10× faster simulation) while maintaining accuracy.

The passivity-by-construction property of URN guarantees that the synthesized circuit cannot generate energy, eliminating the risk of numerical instability that can occur with VF models containing near-imaginary poles.

VI. THEORETICAL ANALYSIS

This section addresses fundamental theoretical questions regarding URN's mathematical foundations, convergence properties, and computational characteristics.

A. MATHEMATICAL STABILITY AND CONVERGENCE

The optimization landscape for time constant parameters τ_i in (10) is non-convex. Unlike Vector Fitting, which guarantees global convergence for fixed poles via linear least squares, URN relies on gradient-based optimization. We analyze convergence behavior through both theoretical arguments and empirical validation.

1) Multi-Start Strategy Justification

URN uses $n_{\text{restarts}} = 8$ independent optimization runs with random initialization. The probability of missing the

TABLE 7. Hyperparameter Sensitivity: Sparsity Weight λ

λ	Error (%)	# Mechanisms	Discovered Types
0.001	1.8	6	Debye(2), CC(2), CPE(2)
0.005	2.1	4	Debye(1), CC(2), Warburg(1)
0.01	**2.8**	**3**	**Ohmic, CC, Warburg**
0.02	3.5	3	Ohmic, CC, Warburg
0.05	5.2	2	Ohmic, CC
0.1	8.7	1	Ohmic

global minimum decreases exponentially with the number of restarts:

$$P(\text{miss}) = (1 - p_{\text{basin}})^{n_{\text{restarts}}} \tag{24}$$

where p_{basin} is the probability of initializing within the attraction basin of the global minimum.

Empirical analysis across 13 benchmark cases shows that different random starts converge to solutions with relative error spread of $<15\%$ (Table 4). The low variance indicates that:

1) The loss landscape has a dominant basin containing near-optimal solutions
2) Multiple local minima, if they exist, have similar objective values
3) The L1 regularization term creates a convex envelope that guides optimization toward sparse solutions

2) Structured Initialization

Rather than purely random initialization, URN uses physically-informed starting points:

$$\tau_i^{(0)} = 10^{\log_{10}(\omega_{\min}^{-1}) + i\cdot\Delta}, \quad \Delta = \frac{\log_{10}(\omega_{\max}^{-1}/\omega_{\min}^{-1})}{M-1} \tag{25}$$

This logarithmic spacing ensures coverage of the measured frequency range and reduces the search space dimension compared to arbitrary initialization.

3) Convergence Guarantee via Regularization

The L1 regularization term $\lambda\|\mathbf{w}\|_1$ ensures that the optimization problem has a unique minimum when $\lambda > 0$ [22]. While the overall problem remains non-convex due to the nonlinear basis functions, the regularization prevents pathological cases where multiple basis functions with different τ values produce identical fits.

B. HYPERPARAMETER SENSITIVITY ANALYSIS

The primary hyperparameter in URN is the sparsity weight λ (denoted `sparsity_weight` in the implementation, default $\lambda = 0.01$). We analyze sensitivity through systematic variation.

1) Sparsity Weight Sweep

Table 7 shows fitting error and discovered mechanisms for the Li-ion battery case across a range of λ values.

The physically correct interpretation (Ohmic + Cole-Cole + Warburg) is stable across $\lambda \in [0.01, 0.02]$. Smaller λ over-fits with redundant bases; larger λ under-fits by eliminating the Warburg tail. This demonstrates that the discovered physics is robust to moderate hyperparameter variation.

2) Cross-Validation Strategy

For production use, we recommend k-fold cross-validation with $k = 5$:

1) Split frequency points into 5 folds
2) Train URN on 4 folds, evaluate on held-out fold
3) Select λ minimizing average validation error

This automated selection removes subjective hyperparameter tuning.

C. COMPUTATIONAL COST ANALYSIS

URN training is more computationally intensive than Vector Fitting. We provide explicit cost comparison and justify the overhead.

1) Time Complexity

- **Vector Fitting**: $O(N_p \cdot K)$ per iteration, where N_p is pole count and K is frequency points. Typically converges in 5–10 iterations. Total: 0.1–1 second.
- **URN**: $O(M \cdot K \cdot E)$, where $M = 29$ bases and $E = 5000$ epochs. With 8 restarts: $\approx$100–150 seconds on CPU.

URN is approximately **100–1000$\times$ slower** than VF for a single fit.

2) Cost-Benefit Analysis

The computational overhead is justified by:

1) **One-time cost**: Impedance characterization is performed once per component; SPICE simulation runs thousands of times during design iteration.
2) **Avoided rework**: VF models with spurious poles require manual inspection and re-fitting. URN's guaranteed stability eliminates this iteration loop.
3) **Physical insight**: Automatic mechanism discovery provides engineering value beyond fitting accuracy.
4) **Passivity guarantee**: VF may require post-hoc passivity enforcement [26], adding complexity and potential accuracy loss.

For batch processing (e.g., parameter sweeps over temperature or bias), GPU acceleration reduces training time to $<$10 seconds per case (future work).

3) Memory Complexity

Both VF and URN have $O(N_p \cdot K)$ or $O(M \cdot K)$ memory requirements for storing basis evaluations. For typical cases ($K \approx 100$, $M = 29$), memory usage is negligible ($<$1 MB).

D. ON "KAN-NESS": TERMINOLOGY CLARIFICATION

A natural question arises: *Is URN truly a "KAN" if the basis functions are fixed physical models rather than learned splines?*

We clarify the relationship between URN and KAN along two dimensions:

1) Shared Philosophy

URN adopts KAN's core insight: **learnable univariate functions on edges** rather than fixed activations on nodes. In standard KAN, these are B-splines with learnable knots. In URN, they are physics-based functions with learnable parameters (τ, α, n).

The key innovation of KAN—that the *form* of the activation function should be learned from data—is preserved. URN simply constrains this learning to circuit-compatible forms.

2) Distinction from Sparse Dictionary Learning

Sparse dictionary learning [27] also uses fixed basis functions with sparse weights. The distinction is:

- **Dictionary learning**: Bases are fixed; only weights are learned
- **URN**: Both basis parameters (τ_i, α_i) *and* weights (w_i) are jointly learned

This joint optimization over continuous parameters is the "KAN-like" aspect. URN learns *which relaxation times exist* and *what exponents describe the distribution*—not just *how much of each pre-defined basis to use*.

3) Physics-Informed KAN

We propose the term **Physics-Informed KAN (PI-KAN)** to describe architectures like URN that:

1) Use learnable basis functions (KAN philosophy)
2) Constrain the function space to physically meaningful forms (physics-informed)
3) Guarantee output realizability (e.g., circuit compatibility)

This parallels the distinction between Physics-Informed Neural Networks (PINNs) [28] and standard neural networks. Just as PINNs embed physical laws as constraints, PI-KAN embeds physical basis functions as the learnable components.

VII. TARGET APPLICATIONS

URN is most effective for problems where (1) the underlying physics involves known relaxation mechanisms, and (2) accurate time-domain simulation is required. We identify three high-impact application domains.

A. WIRELESS POWER TRANSFER (WPT)

WPT systems operating at 6.78 MHz or 13.56 MHz require accurate modeling of:

- **Ferrite core permeability**: Temperature and DC-bias dependent relaxation
- **Litz wire AC resistance**: Skin and proximity effects
- **Compensation network**: Resonant tank tuning

VF struggles with ferrite modeling because the Cole-Cole response appears as a "single broad peak" that VF approximates with multiple sharp poles—leading to spurious ringing during load transients. URN directly identifies the Cole-Cole mechanism with correct α parameter.

Expected impact: 5–10$\times$ improvement in transient simulation accuracy for WPT control loop design.

B. BATTERY MANAGEMENT SYSTEMS (BMS)

Lithium-ion battery impedance exhibits:

- **Charge transfer**: Randles circuit with Warburg diffusion
- **SEI layer**: Constant phase element (CPE) response
- **State-of-charge dependence**: Parameter variation with SOC

URN's automatic mechanism discovery is particularly valuable for BMS because:

1) Different battery chemistries have different dominant mechanisms
2) Aging changes the relative contribution of each mechanism
3) Real-time parameter tracking requires fast, robust fitting

The Warburg element ($n = 0.5$) is automatically detected and synthesized as an RC ladder, enabling accurate prediction of diffusion-limited dynamics during fast charging.

C. EMI FILTER DESIGN

Common-mode (CM) chokes exhibit complex frequency-dependent impedance due to:

- **Winding capacitance**: Creates self-resonance
- **Core loss**: Frequency-dependent permeability
- **Skin effect**: AC resistance increase

Accurate CM choke models are critical for predicting conducted EMI spectra. VF models often produce "phantom resonances" that do not appear in measurements—leading to over-designed or under-performing filters.

URN's pole-free formulation eliminates phantom resonances while capturing the true frequency-dependent behavior through physically-meaningful basis functions.

D. INDUCTION HEATING

Induction heating workpieces exhibit strongly nonlinear, temperature-dependent surface impedance. The skin depth:

$$\delta = \sqrt{\frac{2}{\omega\mu(T)\sigma(T)}} \tag{26}$$

varies with temperature, affecting power distribution.

URN's skin effect basis with the Dowell ladder synthesis [3] provides a SPICE-compatible model that can be coupled with thermal simulation for accurate power prediction.

E. LIMITATIONS AND WHEN NOT TO USE URN

URN is *not* recommended for:

1) **Resonant structures**: Antennas, cavities, and waveguides with true poles

2) **Multi-port S-parameters**: Current single-port limitation
3) **Unknown physics**: Systems with mechanisms not in the 29-function library

For these cases, VF or physics-based modeling remains appropriate.

F. FUTURE WORK

Several extensions are planned for future development:

1) **Multi-port extension**: Generalize URN to handle multi-port S-parameter data using Block Lanczos [25] for coupled inductor and transformer modeling.
2) **GPU acceleration**: Implement batch training on GPU for large-scale parameter sweeps and real-time fitting applications.
3) **Automatic basis expansion**: Develop methods to automatically add new basis functions when existing library cannot adequately fit the data, while maintaining circuit compatibility.
4) **Temperature dependence**: Incorporate temperature-dependent parameters for thermal-electrical co-simulation in power electronics design.

VIII. CONCLUSION

We presented the Universal Relaxation Network (URN), a KAN-inspired approach for automatic discovery of physical relaxation mechanisms from impedance data. The key contributions are:

1) **Pole-free formulation**: Unlike Vector Fitting's pole-residue representation, URN uses physically-grounded basis functions that guarantee stability and prevent spurious resonances in time-domain simulation.
2) **Automatic mechanism discovery**: L1-regularized sparse optimization selects 2–4 dominant mechanisms from a library of 29 basis functions, eliminating the need for expert circuit topology selection.
3) **Direct circuit synthesis**: Each basis maps to an RLC ladder via established methods (Valsa, Charef, Dowell), producing SPICE-compatible netlists with guaranteed positive component values.
4) **Superior accuracy**: Benchmarks on 13 test cases demonstrate 3.22% average maximum error, compared to 63.21% for conservative VF and 368.24% for aggressive VF (with 46% invalid models).

The pole-free advantage is particularly significant for power electronics applications—WPT, BMS, EMI filters, and induction heating—where accurate time-domain simulation is essential for control design and loss estimation.

Current implementation: Series and parallel basis topologies with Cauer ladder synthesis. The framework is extensible to hybrid topologies and multi-port systems via Block Lanczos.

Open-source availability: PyTorch implementation (700 lines), CPU-only, training time 100–150 seconds per case. No GPU required. Source code is available at https://github.com/ksugahar/URN.

Reproducibility: The repository includes sample measurement data files (`data/liion_battery_eis.csv`, `data/mnzn_ferrite_impedance.csv`) and validation scripts (`validate_urn_vs_vf.py`, `demo_spice_timedo` to reproduce all experimental results and time-domain simulations presented in this paper.

URN bridges the gap between data-driven flexibility (KAN philosophy) and physics-based interpretability (circuit-compatible constraints), offering a practical tool for impedance modeling in power electronics design.

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